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Input:

```
#include<iostream>
using namespace std;

class hp
{
    int heap[20], heap1[20], x, n1, i;
public:
    hp()
    { heap[0]=0; heap1[0]=0;
    }
    void getdata();
    void insert1(int heap[],int);
    void upadjust1(int heap[],int);
    void insert2(int heap1[],int);
    void upadjust2(int heap1[],int);
    void minmax();
};

void hp::getdata()
{
    cout<<"\n Enter the no. of students: ";
    cin>>n1;
    cout<<"\n Enter the marks: ";
    for(i=0;i<n1;i++)
    {
        cin>>x;
        insert1(heap,x);
        insert2(heap1,x);
    }
}

void hp::insert1(int heap[20],int x)
{
    int n;
    n=heap[0];
    heap[n+1]=x;
    heap[0]=n+1;

    upadjust1(heap,n+1);
}

void hp::upadjust1(int heap[20],int i)
{
    int temp;
    while(i>1&&heap[i]>heap[i/2])
    {
        temp=heap[i];
        heap[i]=heap[i/2];
        heap[i/2]=temp;
        i=i/2;
    }
}

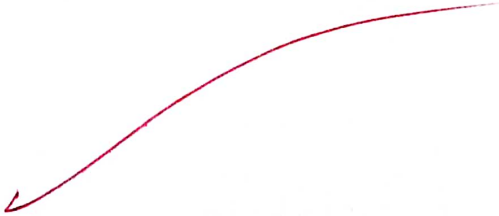
void hp::insert2(int heap1[20],int x)
{
    int n;
    n=heap1[0];
    heap1[n+1]=x;
```

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    heapl[0]=n+1;

    upadjust2(heapl,n+1);
}
void hp::upadjust2(int heapl[20],int i)
{
    int templ;
    while(i>1&&heapl[i]<heapl[i/2])
    {
        templ=heapl[i];
        heapl[i]=heapl[i/2];
        heapl[i/2]=templ;
        i=i/2;
    }
}
void hp::minmax()
{
    cout<<"\n max marks: "<<heapl[1];
    cout<<"\n";
    for(i=1;i<=n1;i++)
    {   cout<<"\n"<<heapl[i]; }
    cout<<"\n min marks: "<<heapl[1];
    cout<<"\n";
    for(i=1;i<=n1;i++)
    {   cout<<"\n"<<heapl[i]; }
}
int main()
{
    hp h;
    h.getdata();
    h.minmax();
    return 0;
}

```



Output:

Enter the no. of students: 5

Enter the marks: 10

24
6
8
9

max marks: 24

24
10
6
8
9

min marks: 6

6
8
10
24
9

